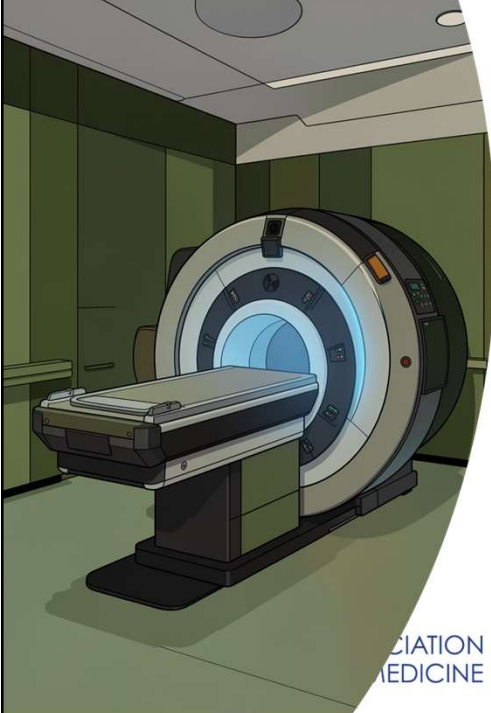




# Robust Adaptive Radiotherapy Planning Approaches

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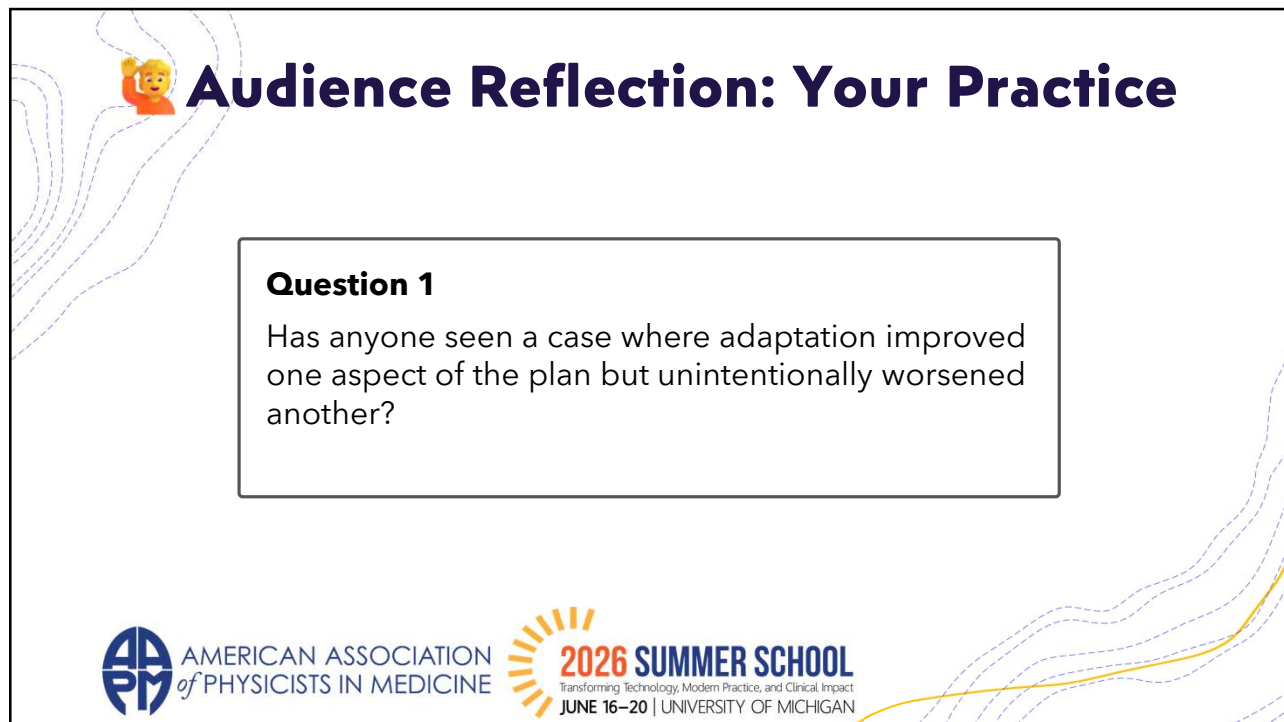
## Disclosures

- None

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# Audience Reflection: Your Practice

**Question 1**

Has anyone seen a case where adaptation improved one aspect of the plan but unintentionally worsened another?

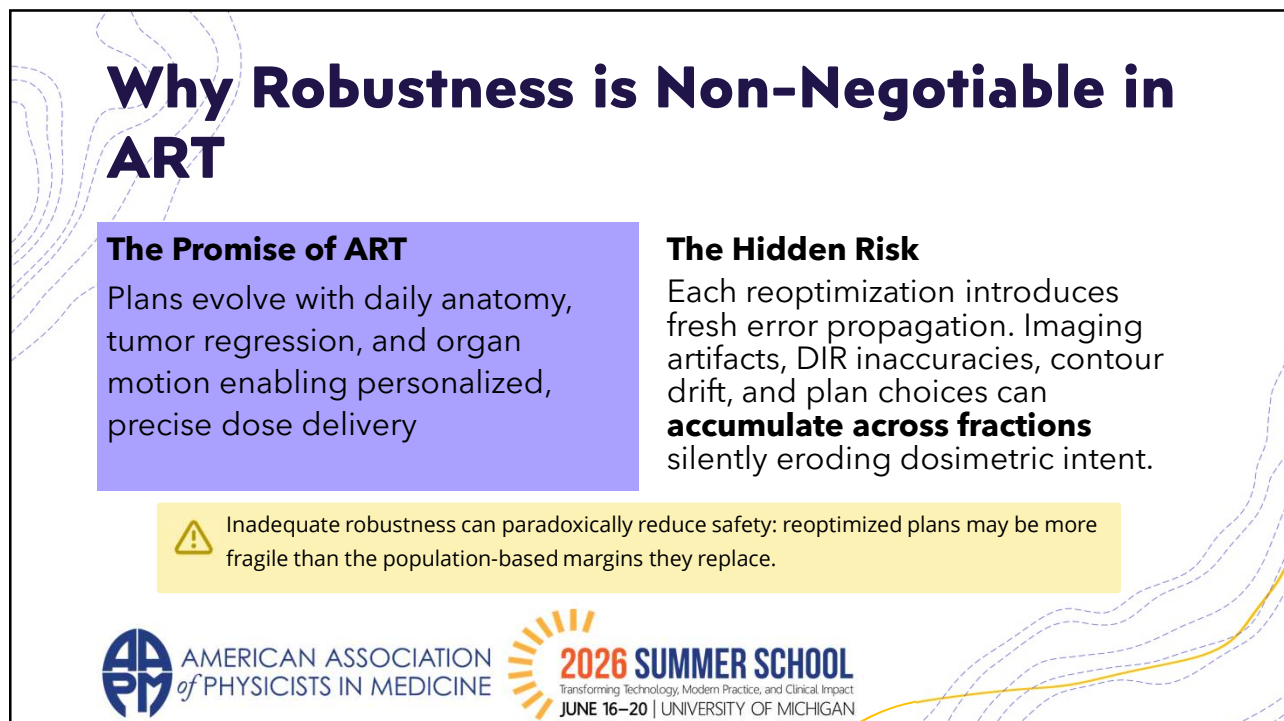


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# Why Robustness is Non-Negotiable in ART

**The Promise of ART**

Plans evolve with daily anatomy, tumor regression, and organ motion enabling personalized, precise dose delivery

**The Hidden Risk**

Each reoptimization introduces fresh error propagation. Imaging artifacts, DIR inaccuracies, contour drift, and plan choices can **accumulate across fractions** silently eroding dosimetric intent.

 Inadequate robustness can paradoxically reduce safety: reoptimized plans may be more fragile than the population-based margins they replace.



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## Audience Working Groups

### Question 2

What are the uncertainties for your table's assigned situation? And which is the most impactful?



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## CBCT Online

### Uncertainty

Autocontour quality on CBCT

Image quality/FOV

DIR accuracy/contour propagation

Dose calculation accuracy

Synthetic CT accuracy

Planning structures position relative  
to target impacting optimization

Intrafraction motion



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## CBCT Offline

### Uncertainty

Autocontour quality on CBCT

Trigger criteria for offline replan

Library plan selection vs full reoptimization

Dose calculation accuracy

Synthetic CT accuracy

Image quality/FOV

Interfraction motion

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## MRI Online

### Uncertainty

Contour uncertainty

Geometric distortion

Electron density assignment

Electron return effect

Limited structures in optimization/Grouped structures weighted equally

Intrafraction motion



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# Proton

## Uncertainty

Contour uncertainty

Setup and range uncertainty

Synthetic CT

Interplay with motion

LET/RBE



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## Audience Working Groups

### Question 3

What are the metrics used to evaluate the uncertainties you listed for your table's assigned situation?



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## CBCT Online

| Uncertainty                                                            | Metric                                                   |
|------------------------------------------------------------------------|----------------------------------------------------------|
| Autocontour quality on CBCT                                            | Dice, Hausdorff distance, etc.                           |
| Image quality/FOV                                                      | Image quality metrics, visual review                     |
| DIR accuracy/contour propagation                                       | Landmark/contour agreement, Jacobian maps                |
| Dose calculation accuracy                                              | CBCT vs CT dose comparison                               |
| Synthetic CT accuracy                                                  | HU error, DVH differences, dose recalculation comparison |
| Planning structures position relative to target impacting optimization | DVH differences, plan deliverability                     |
| Intrafraction motion                                                   | Repeat imaging, target displacement, dose degradation    |

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## CBCT Offline

| Uncertainty                                   | Metric                                                                          |
|-----------------------------------------------|---------------------------------------------------------------------------------|
| Autocontour quality on CBCT                   | Dice, Hausdorff distance, etc.                                                  |
| Trigger criteria for offline replan           | Longitudinal DVH trends                                                         |
| Library plan selection vs full reoptimization | Plan selection DVH vs daily anatomy, comparison to fully re-optimized test plan |
| Dose calculation accuracy                     | CBCT vs CT dose comparison                                                      |
| Synthetic CT accuracy                         | HU error, DVH differences, dose recalculation comparison                        |
| Image quality/FOV                             | Image quality metrics, visual review                                            |
| Interfraction motion                          | Repeat imaging, target displacement, dose degradation                           |

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## MRI Online

| Uncertainty                                                            | Metric                                          |
|------------------------------------------------------------------------|-------------------------------------------------|
| Contour uncertainty                                                    | Dice, Hausdorff distance, etc.                  |
| Geometric distortion                                                   | Phantom measurements, geometric distortion maps |
| Electron density assignment                                            | Dose comparison to CT                           |
| Electron return effect                                                 | DVH of structures adjacent to cavities          |
| Limited structures in optimization/Grouped structures weighted equally | DVH comparisons to original plan, visual review |
| Intrafraction motion                                                   | Cine MRI, target displacement                   |



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## Proton

| Uncertainty                 | Metric                                                   |
|-----------------------------|----------------------------------------------------------|
| Contour uncertainty         | Dice, Hausdorff distance, etc.                           |
| Setup and range uncertainty | Clinical goals evaluated across scenarios                |
| Synthetic CT                | HU error, DVH differences, dose recalculation comparison |
| Interplay with motion       | 4D dose                                                  |
| LET/RBE                     | LET distribution correlation with high-dose regions      |



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## Audience Working Groups

### Question 4

What are mitigation strategies that can be used for the uncertainties you listed for your table's assigned situation?



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## CBCT Online

| Uncertainty                                                            | Metric                                                   | Mitigation Technique                                                           |
|------------------------------------------------------------------------|----------------------------------------------------------|--------------------------------------------------------------------------------|
| Autocontour quality on CBCT                                            | Dice, Hausdorff distance, etc.                           | Physician/physicist review, prioritize high impact OARs, contour action levels |
| Image quality/FOV                                                      | Image quality metrics, visual review                     | Repeat scan possibly with iMAR, higher mAs, breath hold, reposition patient    |
| DIR accuracy/contour propagation                                       | Landmark/contour agreement, Jacobian maps                | DIR QA, limit DIR use to regions with good image quality                       |
| Dose calculation accuracy                                              | CBCT vs CT dose comparison                               | Synthetic CT, HU correction                                                    |
| Synthetic CT accuracy                                                  | HU error, DVH differences, dose recalculation comparison | Site-specific validation, periodic QA                                          |
| Planning structures position relative to target impacting optimization | DVH differences, plan deliverability                     | Creation of planning specific derived structures                               |
| Intrafraction motion                                                   | Repeat imaging, target displacement, dose degradation    | Shorter adaptive workflow, verification CBCT, gating/repositioning             |



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## CBCT Offline

| Uncertainty                                   | Metric                                                                          | Mitigation Technique                                                           |
|-----------------------------------------------|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Autocontour quality on CBCT                   | Dice, Hausdorff distance, etc.                                                  | Physician/physicist review, prioritize high impact OARs, contour action levels |
| Trigger criteria for offline replan           | Longitudinal DVH trends                                                         | Predefined adaptive triggers, scheduled QACT                                   |
| Library plan selection vs full reoptimization | Plan selection DVH vs daily anatomy, comparison to fully re-optimized test plan | Maintain robust plan libraries                                                 |
| Dose calculation accuracy                     | CBCT vs CT dose comparison                                                      | Synthetic CT, HU correction                                                    |
| Synthetic CT accuracy                         | HU error, DVH differences, dose recalculation comparison                        | Site-specific validation, periodic QA                                          |
| Image quality/FOV                             | Image quality metrics, visual review                                            | Repeat scan possibly with iMAR, higher mAs, breath hold, reposition patient    |
| Interfraction motion                          | Repeat imaging, target displacement, dose degradation                           | Shorter adaptive workflow, verification CBCT, gating/repositioning             |

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## MRI Online

| Uncertainty                                                            | Metric                                          | Mitigation Technique                                                        |
|------------------------------------------------------------------------|-------------------------------------------------|-----------------------------------------------------------------------------|
| Contour uncertainty                                                    | Dice, Hausdorff distance, etc.                  | Physician/physicist review                                                  |
| Geometric distortion                                                   | Phantom measurements, geometric distortion maps | Distortion correction algorithms, sequence selection optimized for geometry |
| Electron density assignment                                            | Dose comparison to CT                           | Bulk density assignment or synthetic CT                                     |
| Electron return effect                                                 | DVH of structures adjacent to cavities          | Beam angle selection, robust constraints around ERE zones                   |
| Limited structures in optimization/Grouped structures weighted equally | DVH comparisons to original plan, visual review | Planning structures for close OARs not full OAR                             |
| Intrafraction motion                                                   | Cine MRI, target displacement                   | Real-time gating, beam hold, tracking                                       |

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# Proton

| Uncertainty                 | Metric                                                   | Mitigation Technique                                          |
|-----------------------------|----------------------------------------------------------|---------------------------------------------------------------|
| Contour uncertainty         | Dice, Hausdorff distance, etc.                           | Physician/physicist review                                    |
| Setup and range uncertainty | Clinical goals evaluated across scenarios                | Incorporation of uncertainty into optimization                |
| Synthetic CT                | HU error, DVH differences, dose recalculation comparison | QACT workflow to verify plan                                  |
| Interplay with motion       | 4D dose                                                  | Rescanning/repainting, incorporating motion into optimization |
| LET/RBE                     | LET distribution correlation with high-dose regions      | Beam arrangement to avoid high LET in critical OARs           |



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# Robust Optimization Strategies

| Method                | Pros                 | Cons              | Best fit                                   | Decision cue                                                                                 |
|-----------------------|----------------------|-------------------|--------------------------------------------|----------------------------------------------------------------------------------------------|
| <b>Worst-case</b>     | Maximum safety       | Plan quality loss | High-risk anatomy, tight OAR constraints   | <b>Choose when safety margin matters most.</b>                                               |
| <b>Probabilistic</b>  | Balanced             | Computational     | Well-characterized setup/range uncertainty | <b>Choose when you want balance with quantifiable risk.</b>                                  |
| <b>Multi-scenario</b> | Clinically intuitive | Manual effort     | Workflow-friendly scenario-based planning  | <b>Choose when adaptation must be practical at the console.</b>                              |
| <b>CVaR</b>           | Tail-risk focused    | Complex           | When worst-case outliers drive concern     | <b>Choose when you want to control the worst tail without over-penalizing all scenarios.</b> |



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# How to Handle Change Management

## → Incorporation of Stakeholders is Key

ART requires synchronized decisions across physics, dosimetry, radiation oncology, therapy, and IT

## → Co-Design the Workflow

Build the process *with* stakeholders, not *for* them

## → Pilot and Iterate

Start with a small patient cohort; refine based on measurable bottlenecks

## → Sustained Feedback Loops

Continuous improvement through structured debriefs and data review



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# Core Takeaways

- ✓ **Robustness is a prerequisite** — not an optional safeguard — for realizing the promise of ART. Dynamic reoptimized plans can be more fragile than static margins if uncertainty is not explicitly modeled.

- ⓘ Each ART platform demands **platform-specific robust implementation**: SCT-based stopping power for protons, electron return effect correction for MR-linacs, contour uncertainty quantification for all.

- The **ultimate vision**: a seamless intelligent adaptive workflow where robustness is continuously monitored, updated, and integrated into every treatment decision — precision and resilience unified.



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